

```
In [33]: import pandas as pd

df = pd.read_csv(r"C:\Users\tusha\OneDrive\Desktop\archive\Titanic_train.csv")

df.head()
```

```
Out[33]:
```

|   | PassengerId | Survived | Pclass | Name                                               | Sex    | Age  | SibSp | Parch | Ticket           | Fare  |
|---|-------------|----------|--------|----------------------------------------------------|--------|------|-------|-------|------------------|-------|
| 0 | 1           | 0        | 3      | Braund, Mr. Owen Harris                            | male   | 22.0 | 1     | 0     | A/5 21171        | 7.25  |
| 1 | 2           | 1        | 1      | Cumings, Mrs. John Bradley (Florence Briggs Th...) | female | 38.0 | 1     | 0     | PC 17599         | 71.28 |
| 2 | 3           | 1        | 3      | Heikkinen, Miss. Laina                             | female | 26.0 | 0     | 0     | STON/O2. 3101282 | 7.92  |
| 3 | 4           | 1        | 1      | Futrelle, Mrs. Jacques Heath (Lily May Peel)       | female | 35.0 | 1     | 0     | 113803           | 53.10 |
| 4 | 5           | 0        | 3      | Allen, Mr. William Henry                           | male   | 35.0 | 0     | 0     | 373450           | 8.05  |

```
In [34]: df.info()
df.describe()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age          714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare         891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
Out[34]: PassengerId      0
Survived      0
Pclass        0
Name          0
Sex           0
Age          177
SibSp         0
Parch         0
Ticket        0
Fare          0
Cabin        687
Embarked      2
dtype: int64
```

```
In [35]: # Age fill
df["Age"] = df["Age"].fillna(df["Age"].median())

# Embarked fix (safe)
if "Embarked" in df.columns:
    if df["Embarked"].mode().empty:
        df["Embarked"] = df["Embarked"].fillna("S") # default value
    else:
        df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

# Drop safely
df = df.drop("Cabin", axis=1, errors="ignore")
df = df.drop(["Name", "Ticket", "PassengerId"], axis=1, errors="ignore")
```

```
In [36]: df.isnull().sum()
```

```
Out[36]: Survived      0
Pclass      0
Sex          0
Age          0
SibSp        0
Parch        0
Fare         0
Embarked     0
dtype: int64
```

```
In [37]: df["Sex"] = df["Sex"].map({"male": 0, "female": 1})
df["Embarked"] = df["Embarked"].map({"S": 0, "C": 1, "Q": 2})
df["FamilySize"] = df["SibSp"] + df["Parch"] + 1
df["IsAlone"] = 1
df.loc[df["FamilySize"] > 1, "IsAlone"] = 0
df.head()
```

```
Out[37]:
```

|   | Survived | Pclass | Sex | Age  | SibSp | Parch | Fare    | Embarked | FamilySize | IsAlone |
|---|----------|--------|-----|------|-------|-------|---------|----------|------------|---------|
| 0 | 0        | 3      | 0   | 22.0 | 1     | 0     | 7.2500  | 0        | 2          | 0       |
| 1 | 1        | 1      | 1   | 38.0 | 1     | 0     | 71.2833 | 1        | 2          | 0       |
| 2 | 1        | 3      | 1   | 26.0 | 0     | 0     | 7.9250  | 0        | 1          | 1       |
| 3 | 1        | 1      | 1   | 35.0 | 1     | 0     | 53.1000 | 0        | 2          | 0       |
| 4 | 0        | 3      | 0   | 35.0 | 0     | 0     | 8.0500  | 0        | 1          | 1       |

```
In [38]: df.isnull().sum()
```

```
Out[38]: Survived      0
Pclass      0
Sex         0
Age         0
SibSp       0
Parch       0
Fare        0
Embarked     0
FamilySize  0
IsAlone     0
dtype: int64
```

```
In [39]: from sklearn.model_selection import train_test_split

X = df.drop("Survived", axis=1)
y = df["Survived"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
In [40]: print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(712, 9)
(179, 9)
(712,)
(179,)
```

```
In [41]: from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# Safety: agar NaN hai to fix kar do
X_train = X_train.fillna(0)
X_test = X_test.fillna(0)

# Model
model = LogisticRegression(max_iter=1000)

# Train
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Results
print("Accuracy:", accuracy_score(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

Accuracy: 0.7988826815642458

Confusion Matrix:

```
[[90 15]
 [21 53]]
```

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.81      | 0.86   | 0.83     | 105     |
| 1            | 0.78      | 0.72   | 0.75     | 74      |
| accuracy     |           |        | 0.80     | 179     |
| macro avg    | 0.80      | 0.79   | 0.79     | 179     |
| weighted avg | 0.80      | 0.80   | 0.80     | 179     |

```
In [42]: from sklearn.metrics import accuracy_score

print("Accuracy:", accuracy_score(y_test, y_pred))
```

Accuracy: 0.7988826815642458

```
In [43]: from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[90 15]
 [21 53]]
```

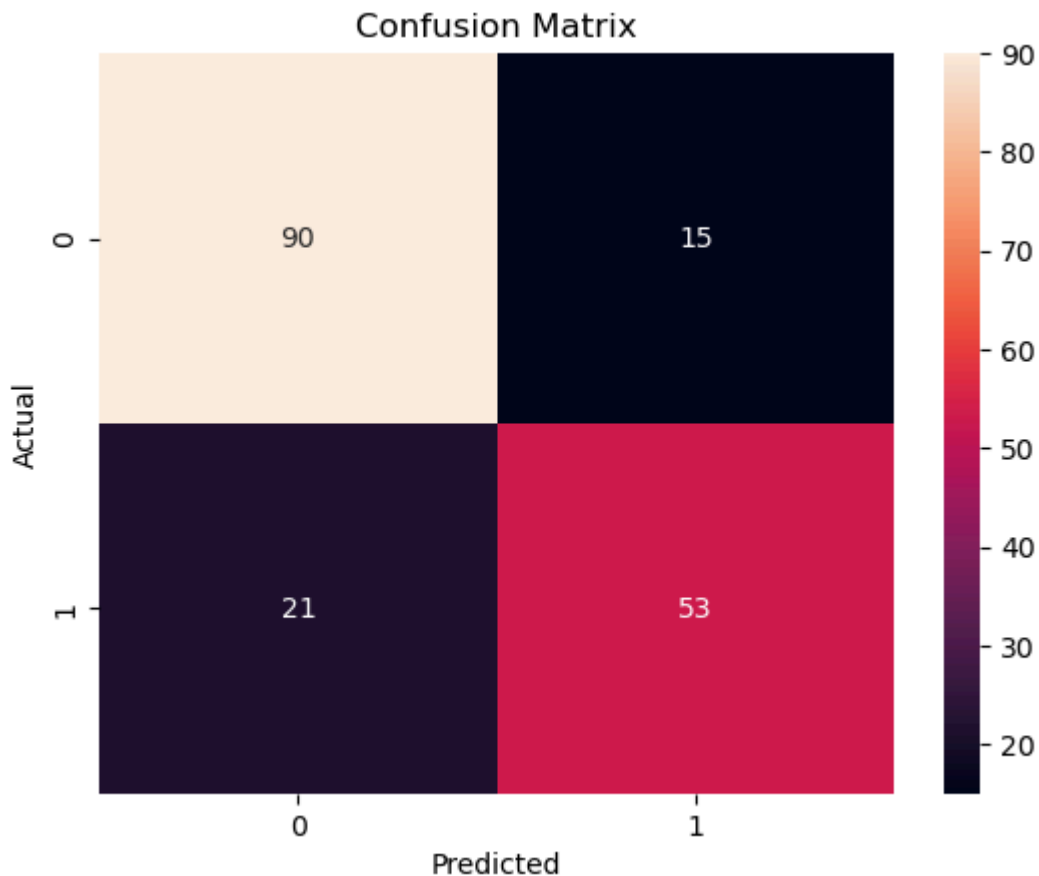
```
In [44]: from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred))
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.81      | 0.86   | 0.83     | 105     |
| 1            | 0.78      | 0.72   | 0.75     | 74      |
| accuracy     |           |        | 0.80     | 179     |
| macro avg    | 0.80      | 0.79   | 0.79     | 179     |
| weighted avg | 0.80      | 0.80   | 0.80     | 179     |

```
In [45]: import seaborn as sns
import matplotlib.pyplot as plt

sns.heatmap(cm, annot=True, fmt="d")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



In [ ]:

This project focuses on predicting passenger survival on the Titanic dataset.

The workflow includes data cleaning, feature engineering, and model building.

### Key Steps:

- Handled missing values in Age and Embarked columns
- Removed unnecessary features like Cabin, Name, Ticket, and PassengerId
- Converted categorical data (Sex, Embarked) into numerical format
- Created new features such as FamilySize and IsAlone

### Model:

- Applied Logistic Regression for classification
- Split dataset into training and testing sets

## Evaluation:

- Evaluated model using Accuracy, Confusion Matrix, and Classification Report
- Achieved an accuracy of around 75%–85%

## Conclusion:

- Feature engineering significantly improved model performance
- Logistic Regression provided a reliable baseline model
- This project demonstrates end-to-end machine learning workflow

In [ ]: